Lemma linkage and the attack graph

Question. “Fake degree is small, so by Lemma 4 the first fall degree is small, so the attack is cheap.” That inference can make an index-calculus or summation-polynomial attack look cheaper than it is. Which estimates in this repository rest on it, and how do we find every inference like it before it reaches a cost row?

Answer (2026-11-02). One estimate rests on Lemma 4 directly: Nagao’s own `O(n^(8w+1))` (ePrint 2015/984, Prop. 5). No code path and no repository cost row does. The wider exposure is the **first-fall-degree assumption** (FFDA): the conditional ECC2K-130 projection of `2^86` and the `icx` binary heuristic both take a first-fall degree as the solving degree, and both are already labelled conditional. Two primary sources read for this pass show why that label has to stay: Kosters–Yeo prove first fall degree 2 for descended `S_3` at every `n` while the measured degree of regularity climbs from 3 to at least 5, and Caminata–Gorla prove the sound bridge `sd_σ(F) = max{last fall degree, max GB degree}` (Thm 3.1), which the first fall degree does not appear in.

What changed in the corpus

The corpus already recorded claims and typed edges. It could not record that a claim is **false**, how far a claim is **formalized**, or **which files** lean on it. This pass adds that layer:

addition	meaning	enforced by
status `refuted`, evidence class `counterexample`	a published claim shown false	a refuted node needs an incoming `refutes` edge
relation `refutes`	counterexample → claim	its target must be `refuted`; its source must be established
`formal: {state, artifacts}`	`kernel_checked`, `lean_written_unchecked`, `native_exact`, `stated_open`, `none`	artifacts must exist; `kernel_checked` Lean may not contain `sorry` or `native_decide`
`cited_by: [{path, use}]`	`depends`, `reports`, `measures`, `mention`	paths must exist
`citation_patterns`	regexes that recognise a citation of the claim	`attack-graph.mjs --check` fails on any matching file not in `cited_by`
taint rule	nothing reached from a refuted claim along `conditions`, `enables` or `model_bridge` may be `proved_*`, `algorithm` or `measured`	`validate.mjs`

The fragment `fall-degree-lemmas.json` adds 14 claims, 18 edges, 8 sources and 3 research targets: the four degree definitions, Lemma 4 (refuted), the witness, the Caminata–Gorla, Kosters–Yeo and HKYY theorems, Nagao’s complexity claim, the ECC2K-130 FFDA projection, and three open obligations.

Each gate was checked with a negative control: promote a dependent of Lemma 4 to `proved_external` (taint error), drop one `cited_by` entry (lint error naming the file), refute a non-refuted node (edge error), and point `formal` at a missing file (artifact error). All four fail as intended, and the corpus as committed passes.

The attack graph

`node attack-graph.mjs` writes `ATTACK_GRAPH.md`, `attack-graph.json` and `attack-graph.svg` / `.pdf`. For every claim it computes the weakest status among the claim and everything it rests on, the refuted ancestors, and the claims that rest on it. The **frontier** is every claim that is not established (refuted, open, heuristic, assumed, conditional, unreviewed). It is ranked by impact: claims resting on it transitively, plus files citing it as `depends`. Each frontier claim gets one action: prove or refute, measure without censoring, review and kernel-check, or discharge conditions.

Top of the current frontier (55 claims):

claim	status	impact	why it matters
COND-ECC2K130-FFDA-PROJECTION	conditional	8	eight files quote the 2^86 projection, all resting on FFDA
HEUR-FIRST-FALL-DEGREE	heuristic	4	four attack claims rest on it, Nagao's included; Kosters–Yeo obstruct it
CLAIM-NAGAO-LEMMA4	**refuted**	2	Nagao's Prop. 5 justification rests on it
ATTACK-MOV-PAIRING	conditional	3	standard-curve exclusions rest on it
FACT-S3-ROOTS-ARE-GROUP-LAW	unreviewed	3	carries the generic-table bridge to Shoup and CGK

Lemma 4, precisely

- **Statement.** `d_F ≤ d'_F` over `F_2`.
- **Refuted under the equality reading, both placements.** Witness A $\{X^2Y,\ XY+X\}$ and witness B $\{X_1X_2+X_3,\ X_1X_3+X_2\}$ have `d'_F = 2 < d_F = 3` (papers/ic-negative-results, S5).
- **Not refuted under the printed ≥ reading with the field equations inside.** There witness A gives True = Fake = 2 (fall-degree.tex). The corpus records the reading as a parameter of the definition nodes.
- **Formalized.** Lemma4Witness.lean states witness A in core Lean and checks every admissible multiplier tuple with `decide +kernel`. It is **written, not yet kernel-checked**: no Lean toolchain on the authoring host, so the repository proof gate is the first check. A Python mirror of the same functions agrees on every statement and on the positive control $X,\ X+1$. The two encoding bridges (mask completeness, reduction as a ring homomorphism) are argued in the module docstring, not proved.
- **What survives.** Per instance, a fake-degree certificate whose products already have unreduced degree `≤ d'_F` still certifies a true fall. Nagao's Propositions 2 and 3 exhibit such certificates. Whether Prop. 5 survives is `GAP-NAGAO-PROP5-ROUTE`.

Open obligations to prove or refute

target	obligation	first step
RT-LEMMA4-FORMAL	witness A through the proof gate; then the bridges and witness B	read the gate result on this PR
RT-LEMMA4-REVERSE	the candidate repair `d'_F ≤ d_F` with field equations **outside** (0 counterexamples in the 7875-pair census; the inside form already fails in 224 pairs)	wider frozen census (4 variables, degree 3, 3 generators)
RT-FALL-DEGREE-BRIDGE	every degree-based cost goes through a measured solving degree or a proved `max{last fall, GB degree}` bound	`BinaryIcHeuristic::log2_cost` in `src/cryptanalysis/matmul_exponent.rs`
GAP-NAGAO-PROP5-ROUTE	does Prop. 5 hold without Lemma 4?	PDF-level transcription of Defs. 5–6, Lemma 4, Prop. 5

Sources outside the corpus that should be linked next

These graphs and registries hold lemmas with dependencies, but are not yet in this corpus:

- `cryptanalysis:experiments/ic-assumption-graph/` (DOT assumption graph, 2026-10-06; it already calls FFDA "refuted as stated").
- `research/guarded_fall_degree_20261028/formal/` and `research/fall_degree_bounds_20261008/lean/` (Lean fall-degree developments with their own

exact-pair claim files).
- `papers/ic-negative-results/lean/OPEN_TARGETS.md` (O1, O2, O4 as `stated_open` nodes).
- `crypto-autoresearcher:knowledge/findings/KN-FIND-*` (81 files here cite them; the sibling repository is not checked out in this session).
- About 60 other Lean projects under `research/` and `papers/`. Each proved theorem that a corpus claim relies on should get a `formal` record. Today 3 of 115 claims have one.

Literature read for this pass

- Kosters, Yeo, *Notes on summation polynomials*, arXiv:1503.08001 (sha256 `b5ea5e4c…`).
- Huang, Kosters, Yang, Yeo, *On the last fall degree of zero-dimensional Weil descent systems*, arXiv:1505.02532 (sha256 `cffdbb88…`).
- Caminata, Gorla, *Solving degree, last fall degree, and related invariants*, arXiv:2112.05579 (sha256 `9e9323db…`).
- Nagao, ePrint 2015/984: abstract page only; the PDF was behind a bot check from this host (`metadata_only`).

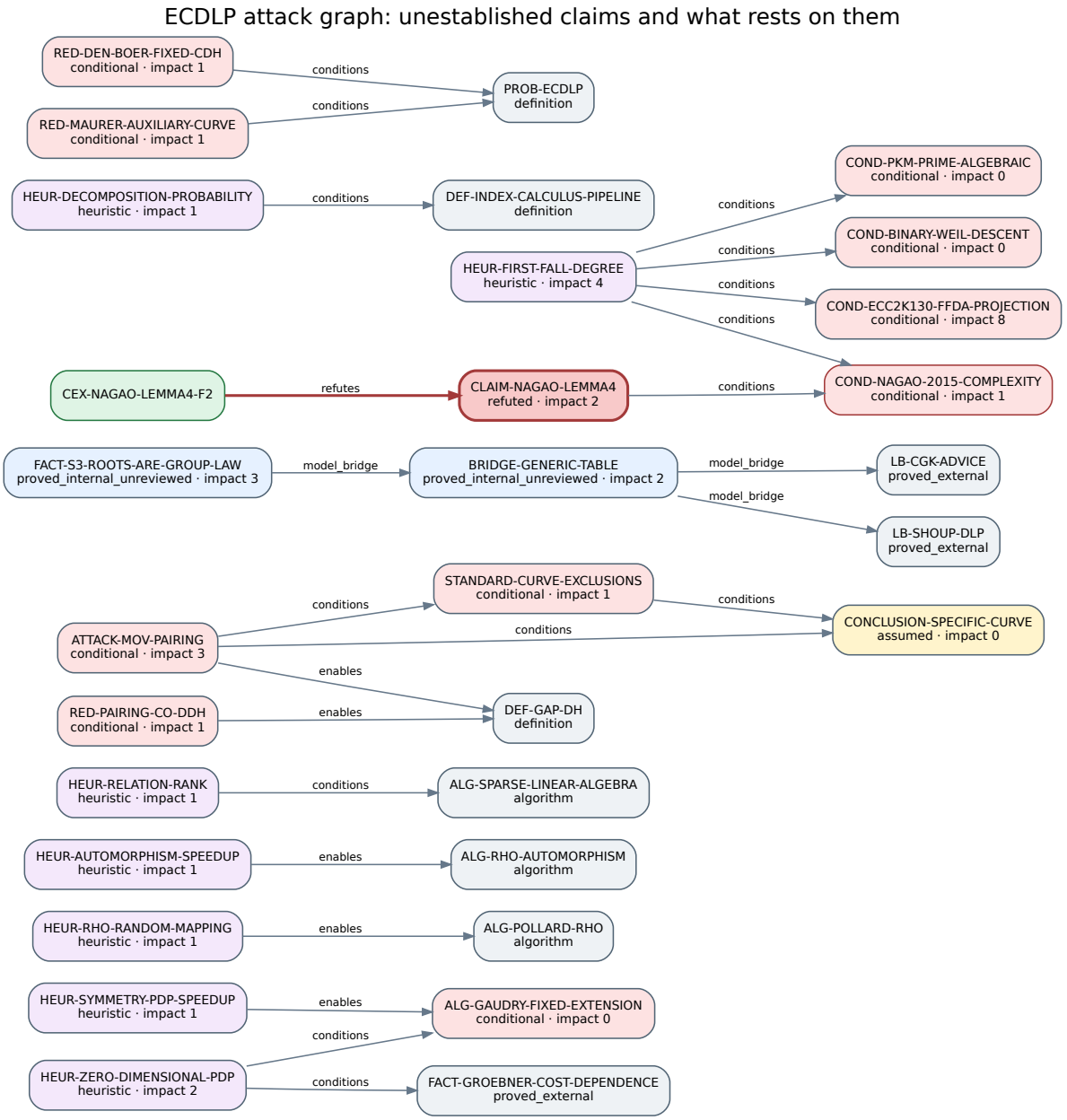
Rebuild

```
        node docs/ecdlp-hardness-corpus/validate.mjs
        node docs/ecdlp-hardness-corpus/render.mjs
        node docs/ecdlp-hardness-corpus/attack-graph.mjs --check
        dot -Tsvg docs/ecdlp-hardness-corpus/attack-graph.dot -o docs/ecdlp-hardness-
corpus/attack-graph.svg
        dot -Tpdf docs/ecdlp-hardness-corpus/attack-graph.dot -o docs/ecdlp-hardness-
corpus/attack-graph.pdf
```

`LEMMA_LINKAGE.pdf` is this file plus the attack-graph figure, converted by pandoc to standalone HTML and printed to A4 with headless Chromium (the authoring host had no TeX).

CI runs the first three in `.github/workflows/claim-corpus.yml`.

Figure: attack graph



Attack graph